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Sexual Selection on Non-Ornamental Traits Is Underpinned by Evidence of Genetic Constraints on Sex-Biased Expression in Dusky Pipefish

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ABSTRACT

Species lacking distinct secondary sex characteristics, such as differences in size or morphology, are often thought to experience lower levels of sex-specific selection in comparison to highly sexually dimorphic organisms. However, monomorphism in classic visible traits could be a result of genetic or physiological constraints that prevent the sexes from reaching divergent fitness optima. Additionally, biochemical and molecular work have revealed a variety of less easily observed phenotypes that nonetheless exhibit profound dimorphism. Sex-specific selection could act on these more subtle, less visible, traits. We investigate sex-specific selection in the polygynandrous dusky pipefish (*Syngnathus floridae*), which lacks distinct secondary sexual characteristics such as size, colour and morphological dimorphism. Using experimental breeding populations, we revealed that although males and females have similar opportunities for sexual selection, only males experience significant sexual selection pressures on body size. We also investigated patterns of sex-biased and sex-specific gene expression in gonads, livers and gills, and tested whether genes with highly divergent expression patterns between the sexes are more likely to be tissue-specific, and therefore relieved of genetic constraints. Sex bias in gene expression was widespread, although the reproductive organs had the most sex-biased and sex-specific genes. Sex-specific selection on gene expression in gills was primarily related to immune response, whereas the liver and gonads had a wide variety of cellular processes, as well as reproductive proteins, showing sex-biased expression. These sex-biased genes showed higher organ-specificity in their expression patterns, suggesting that pleiotropic constraints might have historically impacted the evolution of sex-specific expression patterns. Altogether, we find evidence for ongoing and historical sex-specific selection in the dusky pipefish.

1 | Introduction

A vast number of traits differ between males and females, and these sexual dimorphisms are presumed to have evolved due to selection favouring different trait values in the sexes. Without sexual dimorphism, the population experiences sexual conflict, in which neither sex can maximise their evolutionary fitness (Parker 1979). Decoupling male and female trait values through the evolution of sexual dimorphism resolves sexual conflict over shared traits (Cox and Calsbeek 2009; Mank 2017; Pennell and Morrow 2013; Wright

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et al. 2019). Classic examples of sexual dimorphism tend to revolve around traits easily perceived or measured by humans, such as body size, colour and plumage, conspicuous secondary sexual characteristics for pre-mating sexual selection (Padian 2018; Trail 1990) and some more subtle traits for post-mating sexual selection, such as genital morphology (Evans and Garcia-Gonzalez 2016). Additionally, recent work has revealed extensive dimorphism in molecular, cellular, and biochemical phenotypes (Mank and Rideout 2021), and these invisible phenotypes offer many additional targets of sexual selection.

For sexual dimorphism to evolve, regardless of whether the sex-specific selection is pre- or post-mating, the expression of causal genes needs to be decoupled between the sexes in the relevant cells at the relevant time point during development (Mank 2017; Tosto et al. 2023). Some genes will be constrained and differences in expression between the sexes might be difficult to evolve, for example if those genes have key functions in both males and females across multiple tissue types (Chenoweth, Rundle, and Blows 2008; Mank 2017; Mank et al. 2008; Pavličev and Cheverud 2015; Ruzicka et al. 2019). In these cases, sexual conflict is not easily resolved. In contrast, some genes have more tissue-specific patterns of expression, such as reproductive proteins (Segami et al. 2022), which might experience fewer evolutionary constraints. These tissue-specific genes often have sex-specific functions (Torgerson, Whitty, and Singh 2005) and rapid rates of evolution (Harrison et al. 2015; Swanson and Vacquier 2002) due to either positive selection, potentially to resolve conflict (Swanson and Vacquier 2002), or due to the relaxation of selection (Dapper and Wade 2020; Johnson et al. 2022). Importantly, if pleiotropic constraints prevent the decoupling of genetic architecture, even strong sexually antagonistic selection will be unable to drive the evolution of sexual dimorphism.

Sexual selection is expected to be one of the most powerful forms of sex-specific selection that favours sexual dimorphism (Hedrick and Temeles 1989; Lande 1980, 1981). The strength of sexual selection is linked to mating systems, with the largest difference in fitness variance between males and females occurring when one sex mates multiply but the other does not (polygyny or polyandry) (Wright and Mank 2013). Classic examples of species with sexually selected traits, such as the peacock have these types of mating systems (Petrie, Krupa, and Burke 1999). Species in which both sexes mate multiply (i.e., polygynandry) have less asymmetry in the opportunity for sexual selection between the sexes, which is reflected in less obvious sexual dimorphism associated with pre-mating phases of sexual selection (Jones and Avise 2001). However, polygynandrous species might have a large scope for post-mating sexual selection to shape traits (Birkhead 2010; Wright and Mank 2013), for example in a promiscuous mouse species, traits such as oviduct length and sperm number appear to be under post-copulatory selection (Weber and Fisher 2023). Many reproductive proteins experience relaxed constraints on gene expression (Harrison et al. 2015; Mank et al. 2008; Segami et al. 2022; Torgerson, Whitty, and Singh 2005), further expanding the scope for the evolution of novel post-mating sexual conflict traits to evolve.

In this manuscript, we investigate the relationships between sexual selection, sexual size dimorphism (or the lack thereof), and gene expression, as a form of 'invisible' dimorphism, in the dusky pipefish (*Syngnathus floridae*). Like all syngnathid fishes, *S. floridae* has obligate male parental care in the form of male pregnancy, in which males maintain an osmoregulated environment for the developing embryos in a specialised brood pouch (Ripley 2009). The dusky pipefish is polygynandrous (Jones and Avise 1997b), with variation in mating systems existing across its range (Mobley and Jones 2007, 2009). The species is unornamented, but the ability of males in other syngnathid species to uptake nutrients from aborted embryos (Lin et al. 2024; Sagebakken et al. 2010) indicates a potential scope for sexual conflict, especially if males are able to selectively decrease rates of embryo survivorship, as observed in its congener the Gulf pipefish (*S. scovelli*) (Paczolt and Jones 2010). Any genes experiencing sexually antagonistic selection arising from post-copulatory sexual conflict are most likely autosomal, and found in genomes of both sexes, given that other *Syngnathus* species lack evidence of diverged sex chromosomes (Dubin et al. 2023; Long et al. 2022; Ramesh et al. 2023; Small et al. 2016). As such, any sexual dimorphism, in size or expression, will arise from genetic architecture present in both sexes. All of these characteristics of the dusky pipefish make them an ideal model for answering the following questions: (1) Do the sexes differ in their opportunities for sexual selection at pre- and post-mating episodes, even without visible sexual dimorphism in size or ornamentation? (2) In species without visible sexual dimorphism, do somatic organs exhibit sex bias in the patterns of expression of their genes, relative to reproductive tissues? (3) Is there evidence of patterns reflecting post-mating episodes of selection-shaping patterns in gene expression?

To answer these questions, we established experimental breeding populations of male and female dusky pipefish and used molecular parentage analysis to determine mating success and reproductive success. Using these fitness measures, we then calculated the opportunity for selection and estimated sexual selection on body size. We used RNA-sequencing of gonads, livers, and gills in a subset of individuals to explore patterns of sex-biased gene expression in reproductive and somatic tissues. Importantly, livers play a key role in egg and sperm production in fish (Bruslé and Anadon 2017), so while being somatic tissues, they also have reproductive functions. Our novel combination of quantitative estimates of sexual selection, sex-specific gene expression, and pleiotropic constraints reveals sexual selection acting on invisible variation in this non-ornamented pipefish.

2 | Material and Methods

2.1 | Ethics Statement

Syngnathus floridae were collected in Tampa Bay under special activities licence #SAL 180182-E granted by the State of Florida. All animal use protocols were approved under the University of Tampa AUP #2020-01.

2.2 | Collection of Specimens

Sexually mature female and pregnant male *S. floridae* were sampled between September and December 2020 from three distinct seagrass beds located throughout Tampa Bay, including

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the Saint Joseph Sound (28.0516324, −82.7921436), the southern region of Old Tampa Bay (27.8730210, −82.5397645), and of Fort De Soto (27.6273507, −82.7002240), using 1.3 m wide push-nets with a 6 mm mesh size. Previous studies found little genetic differentiation between pipefish populations along the Gulf Coast of Florida (Flanagan et al. 2016; Mobley et al. 2010), meaning it is likely that pipefish collected from these three nearby locations belong to the same genetic population. Only females > 120 mm in standard length were collected to ensure females were sexually mature, as outlined in previous studies (Mobley and Jones 2007). Pipefish were transported to the University of Tampa in Tampa, FL in aerated 20 L buckets. Once in the laboratory, all pipefish received a freshwater bath lasting for 3 min to remove external parasites and were acclimated to sex-specific holding tanks held at 27°C and a salinity of 25 ppt to mimic their natural habitat.

2.3 | Experimental Breeding Populations

Prior to entering the experiment, pipefish were anaesthetised with a 0.01% clove oil solution for 1 min and marked with three visible implant fluorescent tags (Northwest Marine Technology) along the left side of their dorsal fins to allow for the identification throughout the trials. Protocols for marking followed those outlined in Woods and Martin-Smith (2004). Two colours were used in a combination of eight unique patterns that contained at least one mark of each colour so that all individuals had the same colour combinations but in different locations (i.e., if pipefish had a bias towards one of the colours, all individuals carried that colour). Photographs were taken immediately after marking the fish before placing them back in their original sex-specific holding tanks. A three-day recovery period was given before experimentation.

Methods for the experimental breeding populations were based on a similar study done with *Syngnathus scovelli* (Rose, Paczolt, and Jones 2013). Eleven replicates were completed involving a total of 176 pipefish. Each replicate included eight sexually mature females and eight nonpregnant males housed together in a 140 L tank (77 cm × 30.5 cm × 56 cm) for a total of 14 days. Tanks were equipped with three different sources of biological filtration and an additional filtration source of activated charcoal with 10% water changes performed daily. Pipefish were fed twice daily a mixed diet of day-old artemia, 14-day-old *Artemia* spp. nauplii, live mysids, and various grass shrimp, zooplankton, and copepods collected from the local seagrass beds. Before placement in the trials, all males and females were photographed and weighed. Throughout each replicate, males were checked daily for pregnancies. Once eyespots had developed in at least two-thirds of the pregnant male's brood pouch, the male was euthanised in a 0.05% clove oil solution until ventilation rates dropped to zero for 5 min and dissected. At this stage in the male's pregnancy, any eggs that failed to develop can be easily distinguished from successfully developing embryos (Paczolt and Jones 2010). At the end of the 14-day period, all female and any remaining male pipefish were euthanised and dissected. In all dissections from the mesocosm, the liver, gills and both gonads were removed using aseptic technique and immediately placed in RNAlater and then stored at −80°C until extraction. All remaining tissues were stored in 70% ethanol at 4°C or −20°C.

2.4 | Morphological Analyses

Photographs were analysed in ImageJ v1.53t (Schneider, Rasband, and Eliceiri 2012) to measure standard length, dorso-ventral body depth and snout-vent length (SVL) for each individual. The dorso-ventral body depth was measured from the point in front of the dorsal fin down the bottom of the torso. In males, additional measurements were taken of the brood pouch, including the pouch length, depth, and area. To quantify the degree of sexual dimorphism in *S. floridae* standard length (measured from the tip of the snout to the end of the caudal fin), SVL (measured from the tip of the snout to the urogenital opening), and torso depth were compared between males and females. To account for larger individuals being deeper, torso depth was adjusted for by the standard length of the pipefish prior to analysis. Data were analysed for normality and equal variances and if assumptions were not met nonparametric tests were applied. Student's two sample t-tests were used to detect significant differences between the sexes in standard length, adjusted torso depth, and SVL (with Welch's correction used for SVL only). If significance was present a power test was used to confirm our ability to confidently reject the null.

2.5 | Molecular Parentage Analysis

We used molecular parentage analysis to determine the mating and reproductive success of individuals in our experimental breeding populations. To assign maternity, each male's pouch was dissected following the protocol outlined in Jones and Avise (1997b). Briefly, each pouch was divided into 14 roughly equal parts (once lengthwise and six times horizontally) using a scalpel. From each section, three eggs were randomly selected and placed in a 1.5 mL tube containing 70% ETOH. The position of each egg within the section was recorded. The remaining embryos were counted and placed in a separate 1.5 mL tube of 70% ETOH. With this sampling scheme, up to 42 offspring were genotyped from each full pouch. If the pouch was not completely full, the same sampling scheme was employed for the sections that contained embryos. Brood sizes ranged from 25 to 500 embryos across all pregnant males, so we assessed maternity for 12%–100% of each brood. DNA was extracted from each embryo using a 150 μL Chelex-proteinase K solution (15 mL MilliQ water, 120 μL proteinase K [20 mg/mL], and 0.75 g Chelex). The embryos were incubated in the solution at 56°C for 4 h followed by 100°C for 8 min. For the adults, DNA was extracted from a 5–10 mg piece of the head using the Qiagen Puregene extraction kit.

Four polymorphic microsatellite loci were used to characterise the mating system and genotype of all sexually mature adults and embryos (*micro25.10*, *micro25.22*, *micro11.1*, and *micro22.3*). These microsatellites were previously developed for *S. scovelli* (Jones and Avise 1997a). The markers were amplified using polymerase chain reaction (PCR) conditions modified from Jones and Avise (1997a) and thermocycler conditions outlined in Mobley and Jones (2007). Annealing temperatures of 54°C were used for *micro11.1* and *micro22.3*, and 56°C for *micro25.10* and *micro25.22*. All markers were run for 36 cycles, except for *micro11.1*, which was run for 40 cycles. Primers were labelled with fluorescent dyes and analysed on a 3130xl Genetic

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Analyser at the University of Canterbury, Christchurch, New Zealand using GeneScan 500 LIZ (Applied Biosystems) as the size standard. Alleles were called and sized with the Geneious Prime microsatellite plugin (v1.4.7). Alleles sizes were exported from Geneious and maternity was assigned using Cervus v3.0.7 (Kalinowski, Taper, and Marshall 2007). Cervus computed the number of alleles, observed and expected heterozygosity, and the likelihood ratio (LOD scores) between the offspring and candidate parents by comparing their genotypes. LOD scores necessary to assign parentage with either 95% or 99% confidence were determined by simulating maternity assignments for 100,000 offspring with the following parameters: 8 candidate females, 100% of the candidate females sampled, and the default 1% of loci mistyped. The proportion of typed loci for the simulation was 91.5%. The father for each embryo was known and therefore known fathers were included in the analysis. Cervus assignments were evaluated using trio LOD scores which consider any potential genotyping errors and the genotypes of the known fathers when assigning maternity. All offspring were required to have a minimum of two typed loci to be included in the analysis.

Previous studies have shown that eggs are spatially segregated by maternity in the brood pouches of *S. floridae* (Jones and Avise 1997b). Therefore, with the sampling scheme of the brood pouch outlined above, we can extrapolate from the sub-sampled pouch sections to estimate maternal reproductive success (following Mobley and Jones 2007). If only one maternal genotype was found for the three embryos from a section, all embryos counted in that section were allotted to that female. In the transition sections, in which the sampled embryos belonged to multiple mothers, the reproductive success for each mother was determined by the proportion of genotyped embryos that were assigned genetically to each mother and the total number of embryos in that pouch section. Each female's reproductive success within a brood was then calculated by summing all embryos assigned to each mother across all the pouch sections. Total reproductive fitness for a male was calculated as the number of embryos within his pouch and for females as the sum of all embryos transferred across all mates (where applicable). Each embryo was given a status of developed (containing eyespots) or undeveloped.

2.6 | Quantifying Sexual Selection Pressures

To investigate the strength and direction of selection acting on SVL in pipefish, we calculated selection differentials. The absolute and standardised selection differentials were calculated as the covariance between SVL (measured from the tip of the snout to the urogenital opening) and relative fitness (Lande and Arnold 1983). We decomposed the total selection differential into three successive, multiplicative episodes (Arnold and Wade 1984a, 1984b). These episodes included premating selection (number of mates) and two episodes of postmating selection (number of eggs transferred per mating and the proportion of embryos that developed). Because we used measures of relative fitness (fitness of one individual/mean fitness for all individuals), individuals who did not mate were included in the calculations for premating selection (the first episode) but not for the following episodes of postmating selection. The opportunity for selection was then calculated as the variance in relative fitness and decomposed into the same selection episodes following techniques outlined in Arnold and Wade (1984b). Due to the independent nature of the trials (i.e., the mating/reproductive success of one trial was not impacted by the mating/reproductive success of another trial), fitness decompositions were calculated separately for each trial and then averaged across all trials to get the final measures of selection. Significance between the sexes and episodes of selection was determined by inspecting the 95% confidence intervals. A 95% confidence interval that crossed zero was interpreted as non-significant selection and non-overlapping 95% confidence intervals between different groups (either between the sexes or across the different episodes of selection) highlighted significant differences between the groups.

2.7 | RNA Extraction and Sequencing

RNA was extracted from each organ from five mid-stage pregnant male and five female individuals using the Qiagen RNeasy kit (Table S4). Samples were disrupted in the RLT-lysis buffer (Qiagen) via mortar and pestle and homogenised via the QIAshredder spin column. The whole liver, both gills and both gonads were included in the extraction. When possible, all organs were extracted and sequenced for the same individual. Five samples for each organ within each sex were sequenced, resulting in a total of 30 samples. Total RNA was sent to Michigan State University RTSF Genomics Core, where libraries were prepared using the Illumina Stranded mRNA Library Preparation kit with polyA selection and IDT for Illumina Unique Dual Index adapters. Libraries were tested for quality control using a combination of Qubit dsDNA HS and Agilent 4200 TapeStation HS DNA1000 assays. One female gill sample did not pass QC and was removed from the analysis. The remaining 29 libraries were loaded onto one lane of a NovaSeq 6000 S4 flow cell v1.5. Base calling was done by Illumina Real Time Analysis v3.4.4 and output was demultiplexed and converted to FastQ format with Illumina Bcl2fastq v2.20.0.

2.8 | Sequence Processing and de Novo Transcriptome Assembly

A total of 1.4 billion 150 bp paired-end reads with an average of 48 ± 7.8 million reads per sample were obtained from the MSU RTSF Genomics core (Table S3). Quality of the samples was assessed using FastQC v0.11.9 (Andrews 2010). Initial trimming and filtering were conducted with Trimmomatic v0.39 (Bolger, Lohse, and Usadel 2014) under the following parameters:

ILLUMINACLIP:NexteraPE-PE.fa:2:30:10 HEADCROP:12 LEADING:3 TRAILING:10 SLIDINGWINDOW:4:15 MINLEN:50.

The trimmed sequences were further filtered to remove ribosomal RNA (rRNA) sequences, potential contaminants and sequencing errors. Reads that originated from rRNA were identified and removed via SortmeRNA v4.3.6 (Kopylova, Noé, and Touzet 2012) specifying the smr_v4.3_fast_db.fasta as the reference and only keeping the reads that did not successfully map back to the reference (i.e., the non-rRNA sequences). Potential contaminant sequences were removed with kraken2 v2.1.2 (Wood, Lu, and Langmead 2019) using the bacterial, Archean,

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viral, plant, fungi and protozoan databases as reference. Lastly, Rcorrector v1.0.6 (Song and Florea 2015) employed a k-mer-based error correction method to account for random sequencing errors present in the reads. Following the stringent filtering process, an average of 40 ± 7.2 million reads per sample were retained, yielding a total of 1.15 billion paired-end reads for *S. floridae*. These high-quality reads were pooled and assembled de novo using Trinity v2.15.1 (Grabherr et al. 2011) using default parameters. The resulting transcriptome assembly consisted of 268,936 unique contigs.

2.9 | Evaluation of Assembly and Assembly Thinning

The basic statistics, such as the number of contigs and N50, of the transcriptome assembly were determined using the Trinity accessory scripts 'TrinityStats.pl.' and 'contig_ExN50_statistic.pl.'. Completeness of the assembly was evaluated with Benchmarking Universal Single-Copy Orthologs (BUSCO v5.2.2) (Manni et al. 2021) in the transcriptome mode using the actinopterygii_odb10 lineage dataset. We calculated the expression-informed N50 (ExN50) statistic, which represents the N50 value for the most highly expressed genes that represent x% of the total normalised expression (Sahraeian et al. 2017), using the Trinity script 'abundance_estimates_to_matrix.pl.' with the TMM normalised expression values.

We observed a large number of duplicated contigs (see Section 3) due to the presence of splicing isoforms (Raghavan et al. 2022), so we opted to thin the assembly using 'SuperTranscripts', which collapses unique and common sequence regions among the splicing isoforms into a single linear sequence (i.e., combining all the isoforms for a single gene; Davidson, Hawkins, and Oshlack 2017). While the generated SuperTranscripts may not exist in a real biological context, they allow for assembly thinning to occur without any sequence loss. SuperTranscripts were generated with Trinity's 'Trinity_gene_splicer_modeler.py' accessory script. The new "SuperTranscriptome" was re-assessed for quality with Nx and ExNx statistics and BUSCO as described above and was used for all subsequent analyses.

2.10 | Read Mapping and Identification of Sex-Biased Genes

To obtain expression estimates the reads from each sample were separately quasi-mapped back to the SuperTranscriptome using salmon v10.1.2 (Patro et al. 2017). A mapping-based transcriptome index was built with a chosen k-mer length of 31 prior to quantifying. Salmon's estimate of the number of reads mapping to each transcript was used for transcript-level abundance and tximport v1.28.0 was then used to compute the gene-level estimates from the transcript estimates. On average, 95%–96% of the RNA-Seq reads mapped back to the assembled SuperTranscriptome per tissue.

The read counts were analysed using DESeq2 v1.40.2 (Love, Huber, and Anders 2014) in R to identify sex-biased genes in each organ. Genes with fewer than 10 read counts or greater than one million counts (summed across all samples) were excluded from the analysis. The DESeq2 model used to determine differential expression was counts ~ group, where group reflected the combined sex and organ type for each sample (i.e., MLiver, FLiver, etc.). Filtering criteria and thresholds have been shown to have major effects on the number of genes defined as sex-biased (Ingleby, Flis, and Morrow 2014) so for a conservative estimate of sex-biased genes, they were defined as genes with an adjusted $p < 0.05$ and an absolute log2-fold change of expression (LFC) value greater than 2. To evaluate whether sex-biased genes were sex-biased in multiple organs, the sets of male-biased genes and female-biased genes in each organ were compared with the package UpSetR v1.4.0 (Gehlenborg 2019; Conway, Lex, and Gehlenborg 2017). To more clearly identify patterns across organs, categories of sex-biased expression were determined within each organ based on the male–female LFC distribution with low bias (2–3 LFC), medium bias (3–5 LFC), high bias (5–10 LFC), and extreme bias (> 10 LFC). Sex-specific genes were also identified within each organ if the normalised read counts were less than ten within all samples of one sex and the opposite sex had median read counts of ≥ 20. The cutoff "less than 10" accounted for random sequencing noise that slightly increased the expression of genes that may have otherwise had an expression of zero, and the median was chosen over other possible alternative summary statistics, such as the mean, to account for extreme expression values in the dataset and genes that may show a high degree of variation in expression across the different samples. The upper limit threshold (≥ 20) was determined through a visual investigation of median expression in males and females.

2.11 | Index of Tissue Specificity (τ)

The tissue specificity index (τ) (Yanai et al. 2005) was used to measure tissue specificity in all significantly expressed genes (i.e., genes that were not filtered out due to low counts) for *Syngnathus floridae*. To calculate this index, we used the relative abundance estimator provided by salmon, Transcripts Per Million (TPM). TPM provides the number of transcripts from a particular gene normalised first by gene length, and then by sequencing depth (in millions) in the sample. The TPM estimates were used in the calculation of tissue specificity (τ) for each SuperTranscript with the equation (Mank et al. 2008):

$$\tau = \frac{\sum_i \left[1 - \frac{\ln(\mathrm{TPM}_i)}{\ln(\mathrm{TPM}_{max})}\right]}{N - 1} \tag{1}$$

where N is the number of tissues examined, TPM_{max} is the maximum average TPM for a given tissue type, and TPM_i is the average TPM for tissue i. Because TPM values approaching zero are impacted by sampling stochastically, any genes that had a $\mathrm{TPM}_i < 2$ were set to 2 (Mank et al. 2008).

For the calculation of tissue specificity, the organ category 'Gonads' was divided into 'Ovaries' and 'Testes' to allow for identification of ovary- and testes-specific genes. Additionally, genes that were removed in the differential expression analysis due to counts of less than 10 across all samples, greater than one million, or were categorised as outliers by DESeq2 were also removed from the TPM dataset prior to calculating τ. The range for τ is 0–1, where $\tau = 0$ indicates even expression across all tissues

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and $\tau = 1$ indicates entirely tissue-specific expression. The relationship between τ and sex bias was then determined within each organ using a Spearman correlation. Our estimate for sex bias was the absolute value of the square root transformed logFC to eliminate the sign differences of male- versus female-biased genes and account for the magnitude of differences in gene expression between the sexes.

2.12 | Gene Ontology Analysis of Sex-Biased Genes and Identification of Sex-Specific Genes

To explore the functional aspect of sex-biased genes we used the PANTHER database (v18.0) for gene ontology analysis via a common reference proteome (Mi, Muruganujan, and Thomas 2012; Thomas et al. 2022). We constructed a local BLAST reference database with the zebrafish (*Danio rerio*) proteome that was downloaded from NCBI. All differentially expressed trinity genes were then searched against the zebrafish reference database with BLASTX. BLAST hits were then filtered to ensure only one hit per gene first by lowest e-value and then by highest percent identity. Any identical hits were then removed to ensure only one unique BLAST hit for each transcript. If there was no BLAST hit, the gene was not included in the PANTHER analysis. The list of protein IDs was then uploaded to the PANTHER database where we performed an analysis using the PANTHER Protein Class Database and zebrafish as the reference organism. Results were then recorded as frequency of total hits in each database for every organ separately. Sex-specific genes were blasted against the *Syngnathus scovelli* genome to retrieve gene names for each Trinity gene. The same filtering as explained above was applied to ensure there was only one hit per sex-specific gene.

3 | Results

3.1 | Sexual Dimorphism in the Dusky Pipefish

We found no significant difference in standard length between male and female *S. floridae* (two sample *t*-test: $t_{170} = 0.865$, $p = 0.388$). However, females were significantly deeper than males (two sample *t*-test: $t_{169} = 4.11$, $p < 0.001$) and had a larger SVL (Welch two sample *t*-test: $t_{146.6} = 3.86$, $p < 0.001$) when compared to males. A power calculation suggests the risk of type I error is very low (power = 0.97 and 0.98 for SVL and adjusted torso depth respectively).

3.2 | Sexual Selection and Mating Behaviour in the Dusky Pipefish

Three out of the four microsatellite loci (*micro22.3*, *micro25.10*, and *micro25.22*) revealed high levels of polymorphism and heterozygosity and were used to confidently assign parentage. The last microsatellite locus (*micro11.1*) appeared to be fixed in the population, with one allele of 50 bp identified in all individuals and was therefore not used. The population of *S. floridae* displayed between 19 and 46 alleles per locus with expected heterozygosity ranging from 0.867 to 0.944, matching results found from the use of these microsatellites on wild populations of *S. floridae* (Mobley and Jones 2007).

Out of the eleven trials, seven resulted in at least one successful mating and were included in the analysis. Across those seven trials, 24 of the 56 males mated at least once and 6 of those males had two mates. No males mated more than twice. The average brood size for pregnant males was 253 ± 28.41 embryos. Within the pregnant male's brood size was not correlated with total pouch area (Pearson's correlation: $r = 0.19$, $t_{22} = 0.90$, $p = 0.38$) or pouch length (Pearson's correlation: $r = 0.11$, $t_{22} = 0.52$, $p = 0.61$). Unmated and mated males did not differ in adjusted torso depth (two sample *t*-test: $t_{54} = -1.12$, $p = 0.27$), brood pouch area (two sample *t*-test: $t_{51} = 0.76$, $p = 0.45$), brood pouch length (two sample *t*-test: $t_{51} = 0.54$, $p = 0.59$), or weight (two sample *t*-test: $t_{54} = 1.23$, $p = 0.22$). However, males who mated were significantly smaller in both standard length (two sample *t*-test: $t_{54} = 2.19$, $p = 0.03$) and SVL (Two sample *t*-test: $t_{54} = 2.29$, $p = 0.02$) than males who did not mate.

Across the seven successful trials, 21 out of 56 females mated at least once, with five of those females mating twice and two mating three times. The average number of eggs transferred by a single female (across all matings where applicable) was 289 ± 48.74, with 202 ± 14.69 eggs transferred per mating on average. The number of eggs transferred was not significantly correlated with the female's standard length (Pearson's correlation: $r = -0.21$, $t_{19} = -0.92$, $p = 0.38$), SVL (Pearson's correlation: $r = -0.12$, $t_{19} = -0.92$, $p = 0.62$), however, we found a significant positive correlation between adjusted torso depth and the number of eggs transferred (Pearson's correlation: $r = 0.43$, $t_{19} = 2.07$, $p = 0.05$). Deeper females appeared to deposit more eggs. Consequently, mated females had a significantly higher adjusted torso depth compared to non-mated females (two sample *t*-test: $t_{54} = -2.97$, $p = 0.004$) but we found no significant difference in terms of standard length (two sample *t*-test: $t_{54} = -0.25$, $p = 0.80$) or SVL (two sample *t*-test: $t_{54} = -0.72$, $p = 0.47$).

Both males and females exhibited a significant positive relationship between relative reproductive fitness (total number of eggs for an individual/mean total number of eggs) and relative mating success (total number of mates for an individual/mean mating success), resulting in slopes of Bateman gradients (β_{SS}) significantly greater than zero ($p < 2.2\text{e-}16$ for both) (Figure 1A). The slopes of the Bateman gradients did not significantly differ between the sexes ($\beta = 1.7\text{e-}03 \pm 2.7\text{e-}02$, $p = 0.94$) and there was no significant interaction between mating success and sex ($p = 0.65$). The total opportunity for selection (I) was not significantly different between males (3.20) and females (3.28) (Table 1), even after decomposing I into the three episodes of selection. For both sexes, variance in fitness due to differences in mating success was the most important contributor for the total opportunity for selection, accounting for 69% and 75% of the variance in fitness in males and females respectively. The next most important contributor for males and females was selection arising from the number of eggs transferred per mate (10% of the variance in fitness for males and 8% for females), followed by selection arising from embryo survivorship (2% of the variance in fitness for both sexes).

While the opportunity for selection remains largely similar between the sexes, our results for the selection differentials (s) and the standardised selection differentials (s') show that males and females experience different patterns of selection

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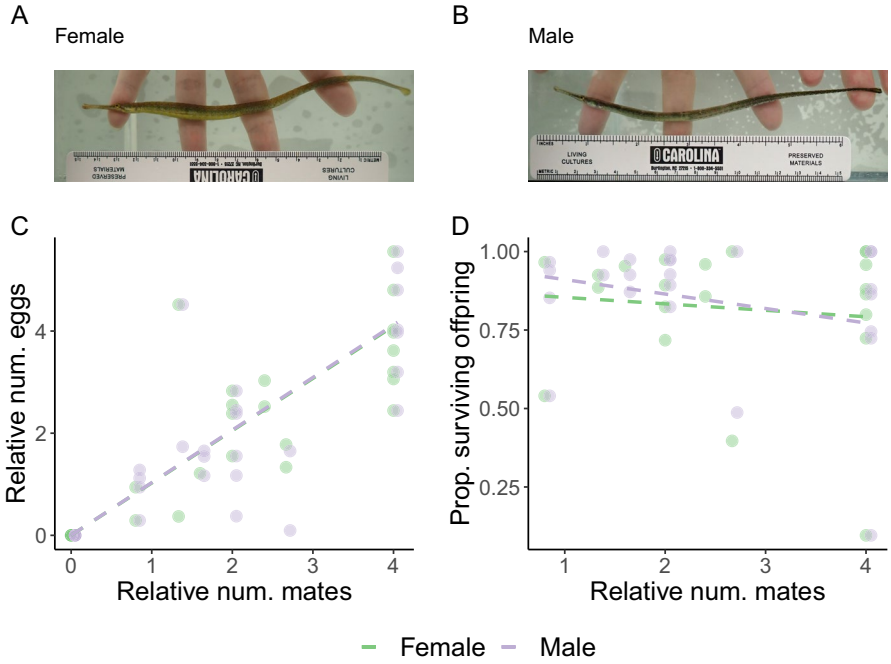

FIGURE 1 | Female (A) and male *Syngnathus floridae* (B) have similar relationships between the relative number of mates (number of mates/mean number of mates) and the relative number of offspring (number of offspring/mean number of offspring). Each additional mating corresponded to an increase in the relative number of embryos for both males and females (C). The proportion of offspring surviving in a brood was not influenced by the relative number of mates in either males or females (D).

TABLE 1 | The opportunity for sexual selection split across mating phases is similar in male and female *Syngnathus floridae*.

Source of variation in fitness	Symbol	Male value (95% CI)	Male (%)	Female value (95% CI)	Female (%)
Premating selection (mating success, w_1)	I_1	**2.20 (1.02, 3.39)**	**68.8**	**2.47 (1.55, 3.40)**	**75.3**
Postmating selection arising from # eggs transferred (eggs per mate, w_2)	I_2	0.32 (−0.173, 0.813)	10.0	0.25 (−0.188, 0.692)	7.7
Variance in number of eggs ($w_1 w_2$)	Subtotal: I_{12}	**2.73 (1.45, 4.01)**	**85.2**	**2.75 (1.84, 3.66)**	**83.7**
Postmating selection arising from embryo survivorship (embryo success, w_3)	I_3	0.0659 (−0.0567, 0.188)	2.06	0.0671 (−0.0556, 0.190)	2.04
Total opportunity for sexual selection ($w_1 w_2 w_3$)	I	**3.20 (1.22, 5.18)**	**100**	**3.28 (1.53, 5.03)**	**100**

Note: Shown are the mean and confidence intervals derived from the estimates of each phase within each trial. The percentages are calculated from the means shown in this table and reflect the fact that most of the variation in fitness arises from premating selection. Bold values reflect selection estimates whose 95% confidence intervals (CI) did not overlap zero.

on body size. Smaller males had higher fitness overall, with the absolute selection differentials being significantly negative in the premating selection and total selection differential (Table 2). This matched our finding that mated males were significantly smaller than unmated males. However, of the males that mated, larger males had a higher fitness in terms of the number of eggs transferred per mate (postmating selection) with a significantly positive absolute selection differential. We did not observe any significant selection differentials for female body size, but the magnitude of the estimates was always opposite the males (Table 2).

3.3 | Patterns of Gene Expression Across Organs

The Trinity assembly showed a high representation of the Actinopterygian core genes (BUSCO quality: C: 94.6% [S: 7.0%, D: 87.6%], F: 2.1%, M: 3.3%, n: 3640) with an N50 of 1307. The high levels of duplication are not uncommon for *de novo* transcriptome assemblies as it stems from the presence of closely related transcripts that represent splicing isoforms (Raghavan et al. 2022). Once the assembly was collapsed into the SuperTranscriptome the duplication seen in the Actinopterygian core genes decreased considerably (BUSCO quality: C: 92.0% [S:

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TABLE 2 | Males and females experience different patterns of sexual selection on body size.

Selection episode	Male *s* (95% CI)	Male *s′* (95% CI)	Female *s* (95% CI)	Female *s′* (95% CI)
Premating selection (mating success)	−2.83 (−6.14, 0.470)	−0.428 (−0.952, 0.0975)	2.28 (−3.76, 8.31)	0.208 (−0.318, 0.735)
Postmating selection (eggs per mate)	**1.99 (0.0234, 3.95)**	0.303 (−0.0400, 0.646)	−2.56 (−8.05, 2.94)	−0.229 (−0.709, 0.251)
Postmating selection (embryo survivorship)	−0.522 (−1.18, 0.141)	−0.0679 (−0.151, 0.0151)	0.970 (−1.34, 3.28)	0.0844 (−0.114, 0.283)
Total selection differential	**−1.37 (−2.67, −0.0597)**	**−0.192 (−0.376, −0.00922)**	0.694 (−2.09, 3.48)	0.0640 (−0.170, 0.298)

Note: Smaller males had higher fitness overall, with absolute selection differentials being significantly negative in premating selection and positive for postmating selection. We did not observe any significant selection coefficients on female body size, but the magnitude of the estimates was opposite that of males. Bold values reflect selection estimates whose 95% confidence intervals (CI) did not overlap zero.

90.2%, D: 1.8%], F: 3.2%, M: 4.8%, n: 3640) while still retaining the high representation of the Actinopterygian core genes. After creating the super-transcripts the N50 increased to 1653 and the final SuperTranscriptome assembly contained 268,936 super-transcripts. After filtering to only retain genes with a total sum of ≥ 10 and < 1,000,000, we had 164,271 SuperTranscripts.

The principal components analysis revealed that much of the variation in expression is driven by differences among the organs (Figure 2A,B). The first axis explains 13.32% of the variance, and the genes with the highest loadings (above 0.02) on that axis were highly expressed in one organ and not another, mostly the liver (Figure 2C). Similarly, the second axis describes 11.4% of the variance and the top-loading genes again have patterns of expression that differ among organs, with most having high expression in the gonads (Figure 2D). The third axis accounts for 7.7% of variation and primarily is associated with differences between male and female expression in the gonads (Figure 2B,E).

3.4 | Differential Expression in the Dusky Pipefish

We identified one male gill sample as a significant outlier in the differential expression analysis and therefore removed it from downstream analyses. As expected, the gonads showed the highest degree of sex bias, with 14% of genes showing differential expression between males and females (Table 3). The liver showed the next highest number of differentially expressed genes (2.5%), followed by the gills (< 1%). In the gonads, more than twice as many genes were male-biased ($n = 9161$) than female-biased ($n = 4097$), but in the liver over three times as many female-biased genes ($n = 634$) than male-biased ($n = 184$) were found. Both sexes had low numbers of sex-biased genes in the gills ($n = 27$ for both). Genes that were biased in one tissue were more commonly not biased in any of the other tissues. The few genes that do show a bias in more than one tissue type often appear to be biased in opposite directions in the different organs. The highest number of genes that are sex-biased in multiple organs are shared between the gonads and the liver (Figure 3C).

The degree of sex bias (measured as the absolute value of the logFC) was higher in the female-biased genes than male-biased genes in both the gills and the liver, but male-biased genes in the gonads were more biased than the female-biased genes (Figure 3A). LogFC was significantly predicted by tissue type (gill, gonad, liver) and bias type (unbiased, male-biased, female-biased) ($R^2 = 0.377$, F (8, 165,927) = 1.26e04, $p < 2.2e\text{-}16$), and each group (i.e., gonad: male-biased, gill: female-biased, etc.) was significantly different in its level of sex bias from the other groups (Table S1). In the gonads and liver, the largest percentage of the sex-biased genes (for both sexes) were in the medium bias category (3–5 logFC), with the fewest genes falling into the extremely biased category (Figure 3B). We found a different pattern in the gills, as the highest percentage of the sex-biased genes were extremely biased (> 10 logFC; Figure 3B).

Most of the sex-biased genes were unable to be classified to a PANTHER protein family. Of the ones that were classified, we saw a high similarity in the families of genes that were sex-biased in the gonads and liver, but not as much in the gills

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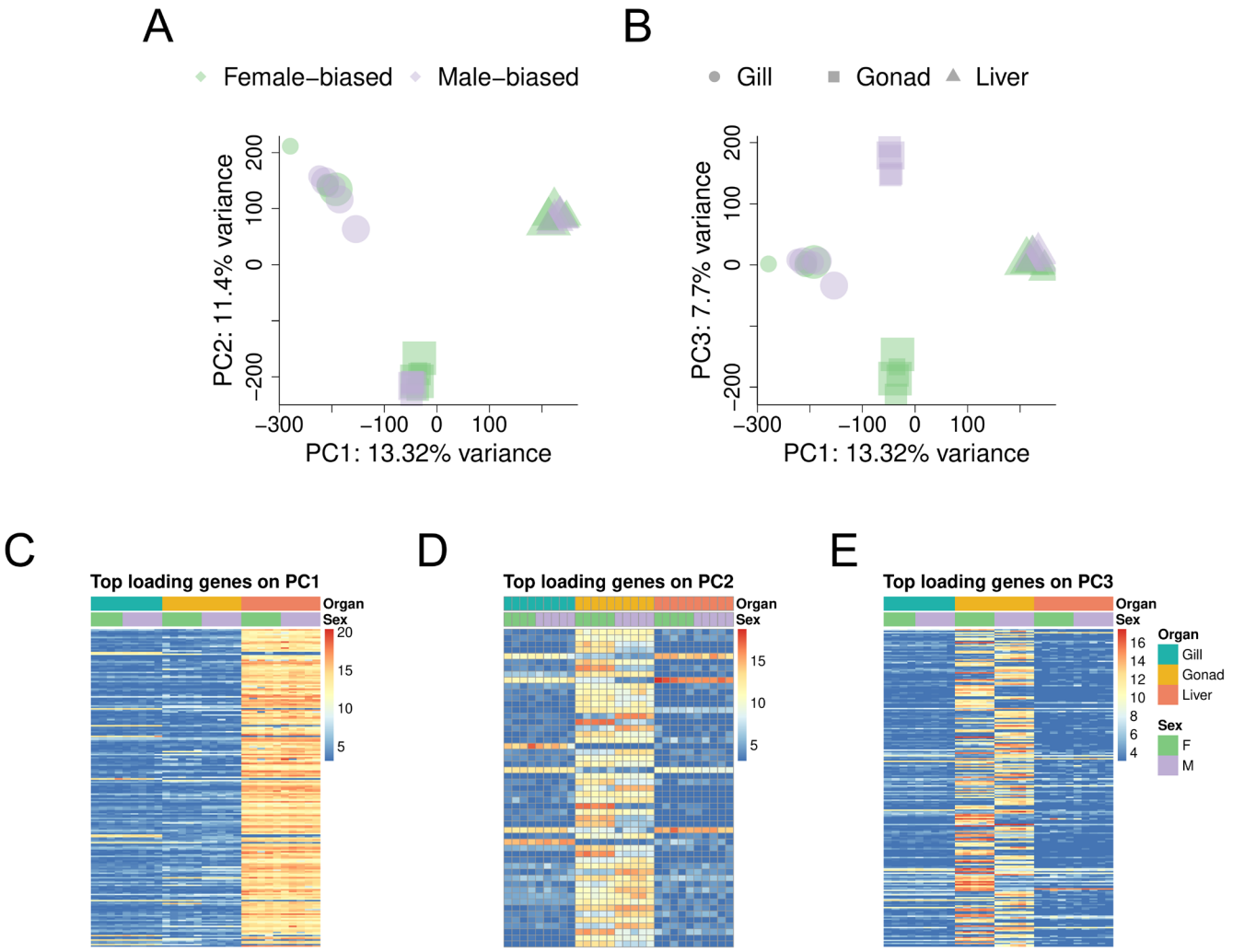

FIGURE 2 | The primary source of variation in the RNA expression in *Syngnathus floridae* is due to differences in expression across organs. The first and second principal components axes differentiate the three tissue types and account for a total of 25% of the variation (A). The third axis accounts for 7.6% of all variation and primarily separates male and female expression in the gonads (B). The top-loading genes on PC1 (with a loading > 0.2) are highly expressed in the liver and to a lesser extent in other tissues (C), whereas the top-loading genes on PC2 are highly expressed in either gills or gonads but generally not both (D), and the top-loading genes on PC3 are sexually dimorphic in expression in gonads but not other organs (E). Points in (A) and (B) are scaled by the mating success of the individual in the experimental breeding populations.

TABLE 3 | Descriptions of sex-biased expression in the different organ types.

		# Genes with a significant sex bias[b]	
Organ	**# Genes with significant expression[a]**	**Female-biased | Male-biased**	**Proportion of all genes that are sex-biased (%)**
Gonads	88,742	4097 | 9161	14.9
Liver	33,320	634 | 184	2.45
Gills	51,867	27 | 27	0.104

[a]The number of genes that were not classified as 'low-counts' or 'outliers' by DESeq2.
[b]Genes were considered sex-biased if they contained an adjusted $p < 0.05$ and a log fold change > |2|.

(Figure 3D). In the liver and gonads, we can see the majority of classified sex-biased genes falling into categories such as actin cytoskeletal muscles, transcription factors, adhesion molecules, regulatory proteins, scaffold/adaptor proteins and voltage-gated ion channels, whereas in the gills we see a large proportion of the sex-biased genes classified as G-protein receptors and the immunoglobulin receptor superfail and cell adhesion molecules.

Similar to sex-biased genes, sex-specific genes had different patterns of sex-specificity across organ types. For the gonads and liver, patterns of sex-specific expression mimicked patterns

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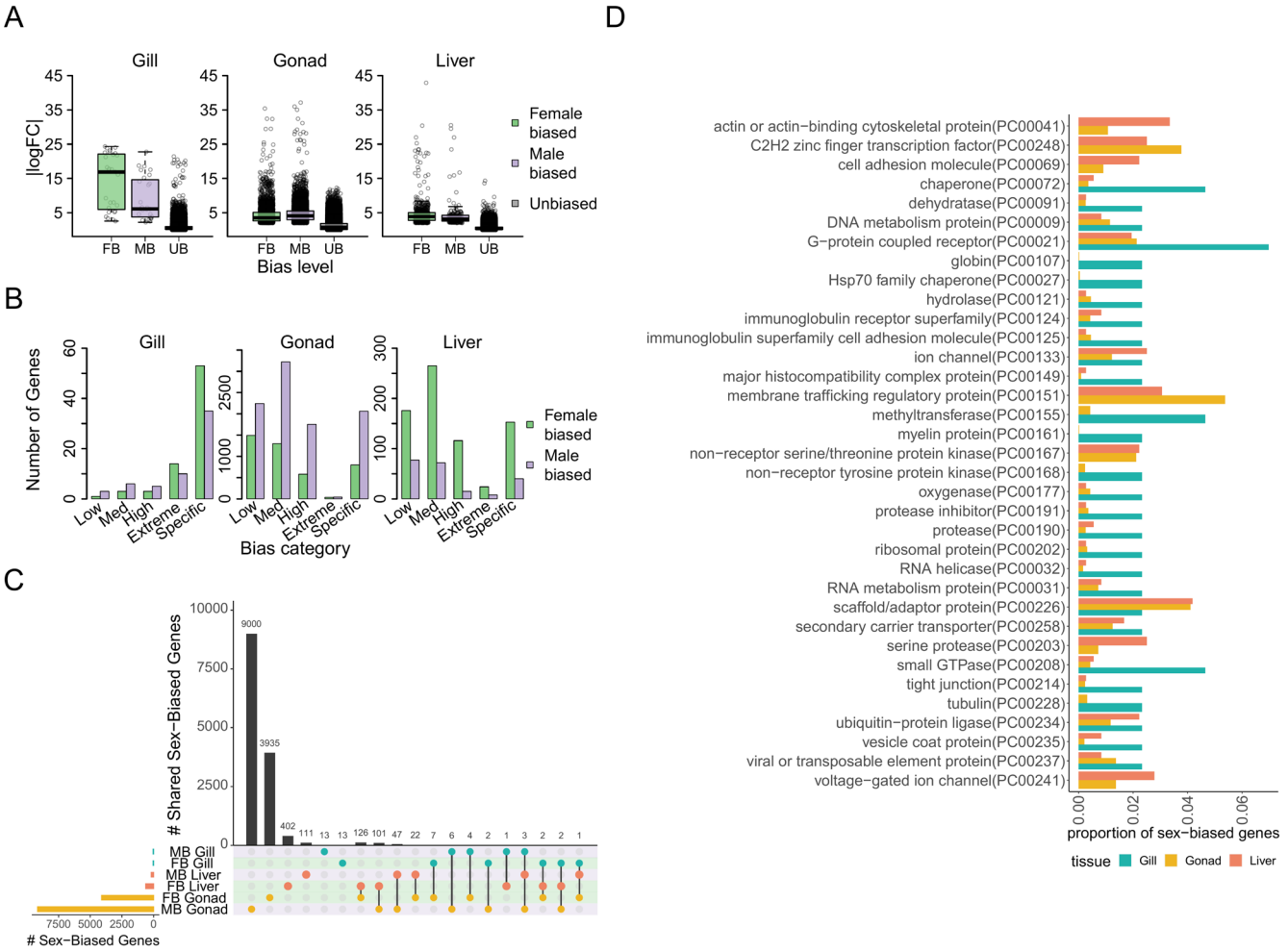

FIGURE 3 | Most sex-biased genes are intermediately biased and are only sex-biased in one organ. Male-biased genes have a higher median log-fold change than female-biased genes in the gonad but not in the gill or liver, although both male- and female-biased genes have some genes with extreme bias (A). The gonad has the highest number of sex-biased genes out of all organs, with more male-biased genes than female-biased (B). Very few genes are sex-biased in multiple tissues, although those that are biased in multiple organs are sometimes biased in opposite directions in different organs (C). Most sex-biased genes were unable to be classified into a PANTHER gene family, but those that were showed similar gene families being sex-biased in gonad and liver but less so for the gill (D).

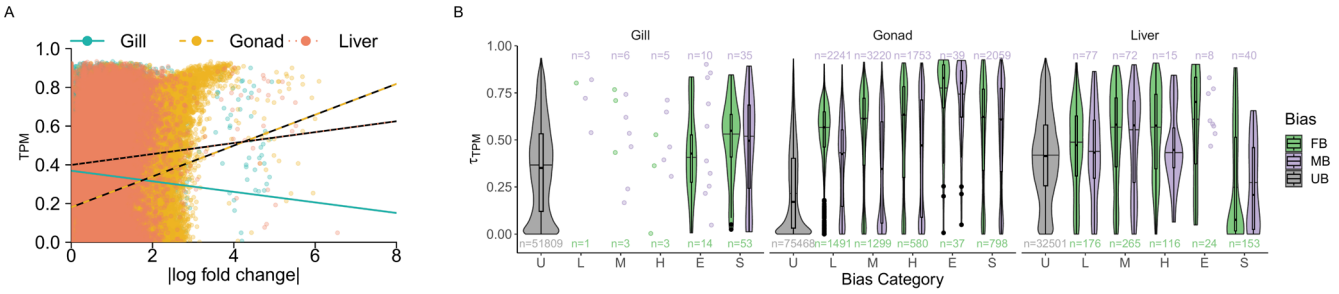

FIGURE 4 | Sex-biased genes in the gonad and liver have higher tissue specificity indices. The three organs have different slopes describing the relationship between sex bias (absolute value of log2 fold change) and tissue specificity (tau calculated from TPM values) (A). In general, the median tissue specificity value is higher in female-biased genes than male-biased genes across categories of sex bias (U = unbiased, L = low sex bias, M = medium sex bias, H = high sex bias, E = extreme sex bias, S = sex-specific) and sex-specific genes demonstrate less tissue specificity than highly or extremely sex-biased genes (B).

of sex-biased expression. The gonads contained the highest number of sex-specific genes, with more male-specific than female-specific genes (F: 798, M: 2059) and the liver had fewer sex-specific genes, but with more female-specific genes than male-specific (F: 153, M: 40). While we again found the lowest number of sex-specific genes in the gill (F: 53, M: 35), the gill showed almost twice as many female-specific genes in comparison to male-specific (Figure 3B). Several of the sex-specific

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genes were identified as genes involved in the reproductive process in the gonad and liver and genes involved related to immune pathways in the gills (Table S1). There were also a few genes that showed sex-specific expression in multiple organs including *claudin-12* and *chromobox protein homologue 2*, which were sex-specific in the gills and gonads, *translationally controlled tumour protein homologue* and *a-kinase anchor protein 9*, which were sex-specific in the gills and liver, and *rho-guanine nucleotide exchange factor 28*, *rho GTPase-activating protein 6*, *purine nucleoside phosphorylase 6*, *transport protein Sec24A*, and *claudin k*, which were all sex-specific in the gonads and liver (Table S1).

3.5 | Tissue Specificity and Sex Bias

The tissue specificity index (τ) was calculated for 161,171 transcripts and ranged from 0 to 0.93. Across all of the organs (gills, gonads and liver), there was an overall and highly significant relationship between sex bias in gene expression and tissue specificity. In the gonads and liver, this was a significant positive correlation ($r = 0.17$ for gonads, $r = 0.055$ for liver; $p < 2.2e\text{-}16$ for both), but in the gills there was a significant negative correlation between τ and sex bias ($r = -0.094$, $p < 2.2e\text{-}16$) (Figure 4A), likely driven by the lower degree of sex bias in that organ (Table 3).

To explore the discrete relationship between sex bias and tissue specificity, we compared measures of τ across the categories of sex bias within each organ (Figure 4B). With few exceptions, female-biased genes appear to be more tissue-specific than male-biased or unbiased genes. In the gonads and liver, we found that the value of τ increases across the categories of sex bias as you move from low-biased genes to extremely biased genes, giving more support to the finding that genes with a higher logFC are more tissue-specific (Figure 4A). This pattern is not as apparent in the gills, but again is likely due to the small number of biased genes in that organ. Interestingly, sex-specific genes in the liver do not appear to be tissue-specific, and in the gonads, sex-specific genes show lower measure of τ than the genes with an extreme sex bias.

An important caveat when interpreting the relationship between sex bias and τ is that genes with high levels of sex bias in one organ are not necessarily highly expressed, or tissue-specific, in that organ. These patterns make interpreting our comparison of tissue specificity and sex bias less straightforward, as a highly sex-biased gene that is also considered tissue-specific might not be tissue-specific in the organ in which it is sex-biased. These misleading genes are only a small subset of the cases, as most sex-biased genes are also highly expressed in the organ in which they are sex-biased (Figure S1), and appear to only occur when a gene is expressed in one or two individuals from one sex and not at all in the other sex in an organ that it is not tissue-specific in. However, these edge cases should be noted because they are not necessarily relieved from pleiotropic constraints. Instead, these sex-biased patterns of expression in some tissue-specific genes have evolved in another organ despite their typically low expression in that second organ—potentially this pattern highlights genes that have been co-opted for a secondary function that is sex-specific in that second organ. Alternatively, these patterns could be the result of bioinformatic artefacts, for example chimeric sequences or combined paralogs that were unable to be identified and resolved in our de novo transcriptome.

4 | Discussion

We set out to understand differences between the sexes in the strength of sexual selection and gene expression in a polygynandrous pipefish that lacks visible sexual dimorphism due to secondary sexual traits. We found significant sexual selection on male body size in our experimental breeding populations, in addition to many male-biased genes in gonad and female-biased genes in liver, potentially indicative of sex-specific selection-shaping gene expression in organs involved in reproduction. The sex-specific patterns of expression were primarily apparent in genes with high tissue specificity, suggesting pleiotropic constraints might prevent the evolution of sexual dimorphism. Altogether, our results support the hypothesis that a lack of sexually dimorphic ornamentation does not necessarily mean a lack of conflict, as we found sexually antagonistic selection might be acting in this species, which could result in sexual conflict, on both morphological features and gene expression.

Specifically, we estimated sexual selection coefficients on body size in opposite directions in males compared to females, although only the male coefficients significantly differed from zero. A significant selection coefficient on body size is somewhat surprising in this species, as *S. floridae* populations are often sexually monomorphic for standard length and evidence of size assortative mating was not previously found (Mobley, Abou Chakra, and Jones 2014). However, some populations of *S. floridae* do show a degree of sexual size dimorphism in standard length (Mobley et al. 2010; Mobley and Jones 2007) and the sexes in this study differed in both SVL and torso depth. The finding of selection favouring smaller males was additionally surprising as evidence from studies on wild *S. floridae* shows that larger males often had both a higher reproductive success and higher mating success compared to smaller males (Jones and Avise 1997b; Mobley et al. 2010; Mobley and Jones 2007). This difference may reflect a true pattern from this population of *S. floridae*, as previous work has shown considerable variation in mating behaviour across distinct populations (Mobley and Jones 2007, 2009), or it may be an artefact resulting from housing the pipefish in the lab. Because part of the courtship ritual for this species involves rising together in the water column (personal observation, similar to vertical swimming identified as a component in *S. scovelli* courtship; Anderson and Jones 2019; Partridge, Boettcher, and Jones 2013), the size of the tanks may have inhibited some males from succeeding. However, larger females were still able to mate successfully, leading us to believe tank size would not be the most likely explanation. Larger males may also require more mating events to fill their brood pouches to capacity (Mobley and Jones 2007), so female reproductive output coupled with the length of the trials could have influenced our estimates of selection. This could mean that previously observed increased fitness in larger males in wild populations (Jones and Avise 1997b; Mobley and Jones 2007) may not represent preference for larger sizes, but instead be a result of the larger pouches requiring more matings to fill. It should be noted, however, that wild populations of *S. floridae* were comparable to our study in average

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reproductive success (range of 180–505; Jones and Avise 1997b; Mobley et al. 2010; Mobley and Jones 2007), average male mating success (~1.5; Mobley et al. 2010; Mobley and Jones 2007), and the maximum number of mates per sex in the same population (2 for males and 3 for females; Mobley et al. 2010).

One possibility is that sexual selection in antagonistic directions is counteracted by other forms of selection, for example viability selection. We investigated trade-offs between sex-specific selection (e.g., sexual selection) and sexually concordant viability selection by examining constraints on the evolution of sex-specific expression, measured by estimating tissue specificity. Our results show that sex-biased genes are generally more organ-specific in their expression, consistent with the hypothesis that pleiotropic genes (i.e., those expressed widely) are constrained in their ability to evolve expression patterns that differ between males and females (Mank 2017; Mank et al. 2008). In addition, sex-specific genes have lower tissue specificity indices than genes that are sex-biased but still expressed in both sexes, a pattern that could reflect resolved intra-locus conflict. Specifically, if a gene exhibits sex-specific expression, it escapes selection in the sex in which it is not expressed, leading to a resolution of conflict over its optimal expression level. This release from conflict could mean that pleiotropy is less constraining and could explain the lower τ in sex-specific genes (Tosto et al. 2023). Taken together, this suggests that there may be potential for sexual conflict resolution for genes with lower constraint in *S. floridae*.

Perceptual bias can influence the way we measure sexual dimorphism by favouring traits we can easily see and quantify, such as body length or size. However, dimorphisms extend to many other phenotypes that are less easily observed, such as cell type abundance or gene expression (Mank and Rideout 2021). We thus also interrogated the transcriptomes of gonads, liver and gills to identify patterns of expression associated with sex-specific selection. We found that the gonads especially, but livers as well, displayed sex-biased patterns of expression, and the genes that were sex-biased in one organ were generally not sex-biased in the others. In the gills, which do not serve a reproductive function, we observed only a small number of differentially expressed genes, but those that were differentially expressed generally were related to immunity, consistent with patterns in other pipefishes (Dubin et al. 2023; Pappert et al. 2024). Genes that had the largest bias in expression while still being expressed in both sexes also tended to be tissue-specific in their expression patterns, suggesting that sex-biased expression might more readily evolve in genes with fewer pleiotropic constraints. Altogether, our results suggest that this sexually monomorphic species seems to be experiencing some nuanced patterns of sexual selection.

We also found that one sex tends to have higher overall expression than the other in a given organ, which could reflect sex-specific functions or a sex-specific genetic architecture of traits experiencing sex-specific selection. In the gonads, more genes were male-biased than female-biased, and the male-biased genes were generally more highly male-biased (i.e., higher log-fold change) than female-biased genes. More male-biased than female-biased expression in gonads has been observed in various stickleback species (Lichilín, Taher, and Böhne 2021), zebrafish (Li et al. 2024), and several species of Galloanserae (Gu et al. 2023; Harrison et al. 2015). However, although male and female gonads arise from the same precursor tissue, comparisons of gonadal gene expression are complicated because male and female gonads are comprised of different cell types. The pattern of increased male-biased expression reverses for the liver, as we identified four times as many female-biased as male-biased genes. Fish livers are known to play an important role in vitellogenesis (Bruslé and Anadon 2017), which may be driving the higher numbers of female-biased genes. However, increased proportions of female-biased genes in the liver are not consistent across other species of fish (Lichilín, Taher, and Böhne 2021). In general, female-biased genes had higher tissue specificity metrics than male-biased genes, regardless of the organ or category of fold-change (Figure 4). Possibly genes with female-biased expression are somewhat more constrained in their expression, such that they require higher degrees of tissue specificity to reach high levels of sex-bias. If male-biased genes are less constrained, it could be related to relaxed selection on male syngnathid reproductive genes (Johnson et al. 2022).

Another factor that could impact these patterns is the lack of a known sex determination system in these pipefish. Within syngnathids, variation in sex determination systems exist, with evidence for a single male-specific sex-determining locus in seadragons and some pipefish (Qu et al. 2021), a genomic region consistent with XY patterns of inheritance in a seahorse (Long et al. 2022), and no obvious genetic region associating with sex in *Syngnathus scovelli* (Ramesh et al. 2023; Small et al. 2016) or *Syngnathus typhle* (Dubin et al. 2023). Due to the close phylogenetic relationship between *S. floridae* and other *Syngnathus* species (Hamilton et al. 2017), we view it is unlikely that a sex-specific region of the genome exists. However, if one does exist, it remains unknown and therefore cannot be removed from our de novo transcriptome assembly. If a sex-specific region does exist, its inclusion in the de novo transcriptome could result in misleading patterns of sex-biased or sex-specific expression.

Our RNA-sequencing data reveals patterns which may suggest that sex-specific selection differs among organs. Potentially, sex-specific selection experienced by *S. floridae* is more pronounced on males in the gonads while being higher in females in the liver, if the direction of sex bias reflects the strength of sex-specific selection. However, this assumes that sex-specific selection results in higher expression in the selected sex, but the link between selection and increased sex-specific expression is not always clear. These patterns of differential expression may simply be a result of the different biological function of these organs in males and females (such as vitellogenesis in females, which is absent in males; Rose, Flanagan, and Jones 2015) or different metabolic demands between sexes (e.g., Masonjones 2001). A previous study with a closely related species of pipefish, *Synganthus scovelli*, highlighted the importance of hormone regulation on gene expression as males mimicked female expression more closely after exposure to oestrogen (Rose, Flanagan, and Jones 2015). Achieving sex-specific functions might not require vastly different patterns of expression, however—for example, seminal fluid proteins are highly expressed in male reproductive tissues as well as in the female reproductive tract in *Drosophila melanogaster* (Thayer et al. 2024). To clarify the link between selection and expression patterns, future work will need to study the traits

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under sex-specific selection and their underlying gene expression networks in more detail, perhaps at a more granular (i.e., single-cell) level.

Many of the sex-biased genes in the liver and gonad are in the categories of actin cytoskeletal proteins, transcription factors, cell adhesion, regulation of membrane trafficking, protein kinase and protease, scaffold/adaptor protein and voltage-gated ion channels (Figure 3). These genes suggest widespread differences in cellular function between males and females in these organs. Sex-biased genes in the gill, although there are few of them, include families such as the immunoglobulin receptor superfamily and its adhesion molecules, the major histocompatibility complex proteins, and other roles that are likely immune-related. Several sex-specific genes reinforce this pattern of sex differences in immune function in the gill, such as *vascular cell adhesion protein 1* and *galactose-specific lectin nattectin*, which are male-specific in both gonads and gills and have been shown in zebrafish and multiple finfish to be related to immune responses (Elumalai et al. 2019; Hsieh et al. 2018). Differences in immune function between males and females have previously been demonstrated in pipefish (Dubin et al. 2023; Pappert et al. 2024; Roth et al. 2011), and the male-specific patterns of expression we observed could support the hypothesis the pregnant male pipefish experience different selection pressures with regard to immunity compared to female and non-pregnant pipefish (Parker et al. 2023; Roth et al. 2020).

If we presume that sex-specific expression patterns are reflective of processes experiencing sex-specific selection, proteins related to reproduction also appear to be under sex-specific selection. Genes related to egg production and maturation demonstrate female-specific expression, such as *zona pellucida* (Wassarman 2008), which is female-specific in the gonads, and *vitellogenin 3*, a female-specific gene in the liver. However, genes involved in the development and movement of sperm, such as *spermatogenesis-associated serine-rich protein 2* and *sperm acrosome membrane-associated protein 4*, exhibit male-specific expression in the gonads (Table S1). In addition, two genes that have previously been shown to be responsive to androgens have sex-specific expression: *extracellular serine threonine protein kinase fam 20c* was female-specific in the liver and has been previously identified as highly female-biased in *S. scovelli* livers and up-regulated in males exposed to synthetic oestrogen (Rose, Flanagan, and Jones 2015), and the *androgen-dependent TFPI-regulating protein* which was male-specific in the gonads.

Several sex-specific genes in the gonads have direct links to reproduction, in particular with the fusion of gametes. For example, the *izumo sperm-egg fusion protein 1* and *hatching enzyme 1.2-like* genes had male-specific expression in the gonads and play a role in reproduction and embryo development. Interestingly, *hatching enzyme 1.2-like* was differentially expressed in the brood pouches of pregnant and non-pregnant male *S. scovelli* (Rose et al. 2023). Its sex-specific expression in the gonads could imply that the testis works in tandem with the brood pouch to regulate embryo development. This hypothesis could be supported as well by the male-specific expression of *parvalbumin beta-like* in the gonads, as it was one of the most highly expressed genes in brood pouches of males regardless of reproductive status (Rose et al. 2023). In females, *zonadhesion*, *zona pellucida sperm-binding protein 4*, *zona pellucida protein AX1*, and *zona pellucida glycoprotein 3f* were sex-specific in the gonads. The *zona pellucida* genes regulate the binding of sperm to ovulated eggs (Wassarman 2008), and *zonadhesion* in mammals is testis-specific and encodes the protein related to sperm adhesion to eggs (Hunt et al. 2005). Previous authors have posited that *zonadhesion* evolved a reproductive role in the mammalian lineage (Hunt et al. 2005). If this hypothesis is correct, then pipefish may have convergently evolved a reproductive role for this gene, analagous to the convergent evolution of pregnancy-related genes (Whittington et al. 2015). Presuming that these six genes with male- and female-specific expression patterns in the gonads retain the functions described in other species, these genes could be key candidates for further investigation into sexual conflict and molecular evolution. These genes regulate the direct interactions between gametes of the two sexes and the development of eggs—both of which are the most likely times for post-mating sexual conflict to arise (Paczolt and Jones 2010; Ripley and Foran 2009)—and their sex-specific expression could provide the opportunity for them to evolve to benefit one sex without constraints due to their expression in the other.

Some sex-specific genes were expressed in ways that are different from patterns seen in other syngnathid fishes. The gene *glyceraldehyde-3-phosphate dehydrogenase* was found as a female-specific gene in the gonads of *S. floridae*, but in the closely related species *S. scovelli* this gene was identified to be highly expressed across both pregnant and non-pregnant males in the brood pouch and the skin of both sexes but did not show significant patterns of differential expression (Anderson et al. 2020; Rose et al. 2023). In other syngnathids, the *low choriolytic enzyme-like* gene appears to play an important role in embryo development, given its upregulation in the brood pouches of pregnant male *S. scovelli* (Rose et al. 2023) and a similar gene (*ASTL*) in the brood pouches of mid- and late-pregnancy *Hippocampus abdominalis* (Whittington et al. 2015). However, in *S. floridae*, this gene showed a male-specific expression pattern in the gills, which do not have a reproductive function. The gene *cathepsin K* had sex-specific alleles in *S. typhle* (Dubin et al. 2023) showed female-specific expression in the gonads in our dataset. Additionally, the gene *leukocyte cell-derived chemotaxin-2*, which has previously been associated with bacterial resistance in fish and was one of the most highly upregulated genes at all stages of pregnancy in *H. abdominalis* (Whittington et al. 2015), was identified as a female-specific gene in the gonads of *S. floridae*. The explanation for these different patterns of expression in *S. floridae* compared to other syngnathids is not obvious and could be an artefact of our de novo transcriptome assembly, which contains about ten times more genes than are expected to be found in the genome (Ramesh et al. 2023; Small et al. 2016). This known drawback of de novo transcriptomes is due to bioinformatic challenges resolving true differences between gene sequences, alternatively spliced genes, and misassembled reads (Martin and Wang 2011; Raghavan et al. 2022), and could mean that our annotations are incorrect or are not actually orthologs of those annotated in reference genomes. Alternatively, it is possible that the genes could result from different selection pressures on these tissues and genes in the different species or that these genes have multiple functions in different tissues and are indeed sex-specific in our study's organs while also highly expressed in other organs in the opposite sex.

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Here, we have revealed evidence of ongoing sex-specific selection on the shared trait of body size, patterns of gene expression consistent with pleiotropic constraints factoring into the evolution of sex-biased expression in reproductive processes, and evidence for the resolution of sexual conflict by the evolution of sex-specific patterns of gene expression. These results highlight that even in monomorphic, unornamented species, which are often thought to experience less sexual selection (Harrison et al. 2015; Jones and Avise 2001; Wright and Mank 2013), sex-specific selection and sexual conflict can still occur, emphasising the pervasive nature of sexual conflict (Tosto et al. 2023).

Author Contributions

Nicole M. Tosto: conceptualization, methodology, investigation, and analysis, data curation and visualization, writing. **Emily Rose:** supervision, resources, writing – review and editing. **Heather D. Mason:** supervision, resources, writing – review and editing. **Judith E. Mank:** conceptualization, writing – review and editing, funding acquisition. **Sarah P. Flanagan:** conceptualization, supervision, project administration, funding acquisition, writing, data curation, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

RNA-Seq data and sample metadata: Raw sequence reads from the RNA-sequencing data and sample meta-data are deposited in the NCBI Short Read Archive (BioProject PRJNA1062454; Tosto et al. 2024b).

Parentage analysis: Microsatellite genotyping data are archived with the data on individual traits and reproductive success on zenodo/dryad (Tosto et al. 2024a).

Code: All codes are on GitHub (https://github.com/flanagan-lab/pipefish_sbge), and the version of the repository at the time of submission is archived on zenodo (Tosto, Beasley, and Flanagan 2024; DOI: https://zenodo.org/doi/10.5281/zenodo.10575377). The repository contains the data required to create the figures presented in this manuscript.

Benefits generated: Benefits from this research accrue from the sharing of our data and results on public databases as described above.

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